

Which of the following molecules exhibit an octahedral bonding configuration?

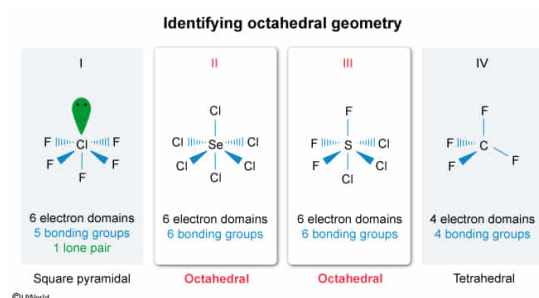
- I. ClF_5
- II. SeCl_6
- III. SF_3Cl_3
- IV. CF_4

- ☐ A. I and II only [11%]
- ✓ ☒ B. II and III only [83%]
- ☐ C. I, II, and IV only [3%]
- ☐ D. II, III, and IV only [2%]

Correct

83% answered correctly

Explanation:



The geometric arrangement of atoms in a molecule can be predicted from its [Lewis structure](#) using the [valence-shell electron-pair repulsion \(VSEPR\)](#) model. According to the model, the electron domains ([bonding](#) and [nonbonding](#) pairs of [valence electrons](#)) located around the central atom of the molecule repel each other, causing them to rearrange so that repulsive forces are minimized.

The **electron geometry** of a molecule describes the arrangement of electron domains whereas the [molecular geometry](#) describes the arrangement of bonded atoms. The molecular geometry depends on the number of bonding versus nonbonding (lone pair) electron domains. Lone pairs exert more repulsion than bonding electrons, resulting in a compression of bond angles relative to the electron geometry. As such, if two molecules have the same number of electron domains but differ in the number of lone pairs, they will have different molecular geometries.

An [octahedral](#) molecular geometry results from six bonding domains and no lone pairs symmetrically arranged around a central atom. The Lewis structures for SeCl_6 and SF_3Cl_3 consist of six electron domains (six single bonds) located on the central atoms, Se and S, respectively, and are octahedral (**Numbers II and III**).

(Number I) When evaluating the *molecular geometry*, lone pair electrons are "invisible" but do exert a force. Although ClF_5 contains six electron groups (five single bonds and one lone pair),

the increased repulsion from the lone pair pushes the five bonding domains closer together to form a **square pyramidal** molecular geometry.

(Number IV) CF_4 is **tetrahedral** with four electron domains (four single bonds) symmetrically arranged around the central C atom.

Educational objective:

The valence-shell electron-pair repulsion (VSEPR) model can be used to predict molecular geometry. An octahedral geometry forms when six bonding electron domains are arranged symmetrically around the central atom of a molecule.

Time spent:0 sec

Subject:
General Chemistry

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System:
Interactions of Chemical Substances